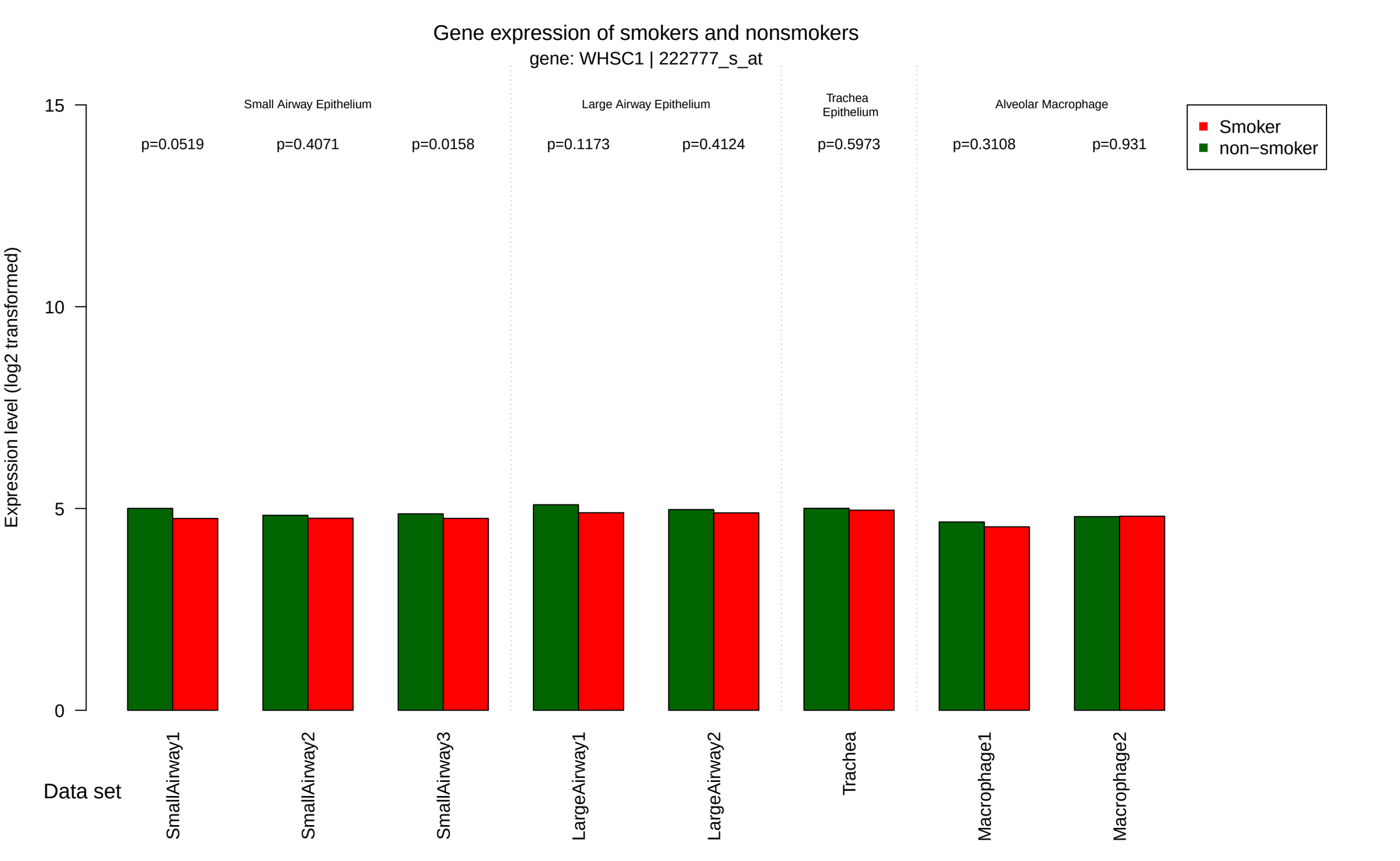
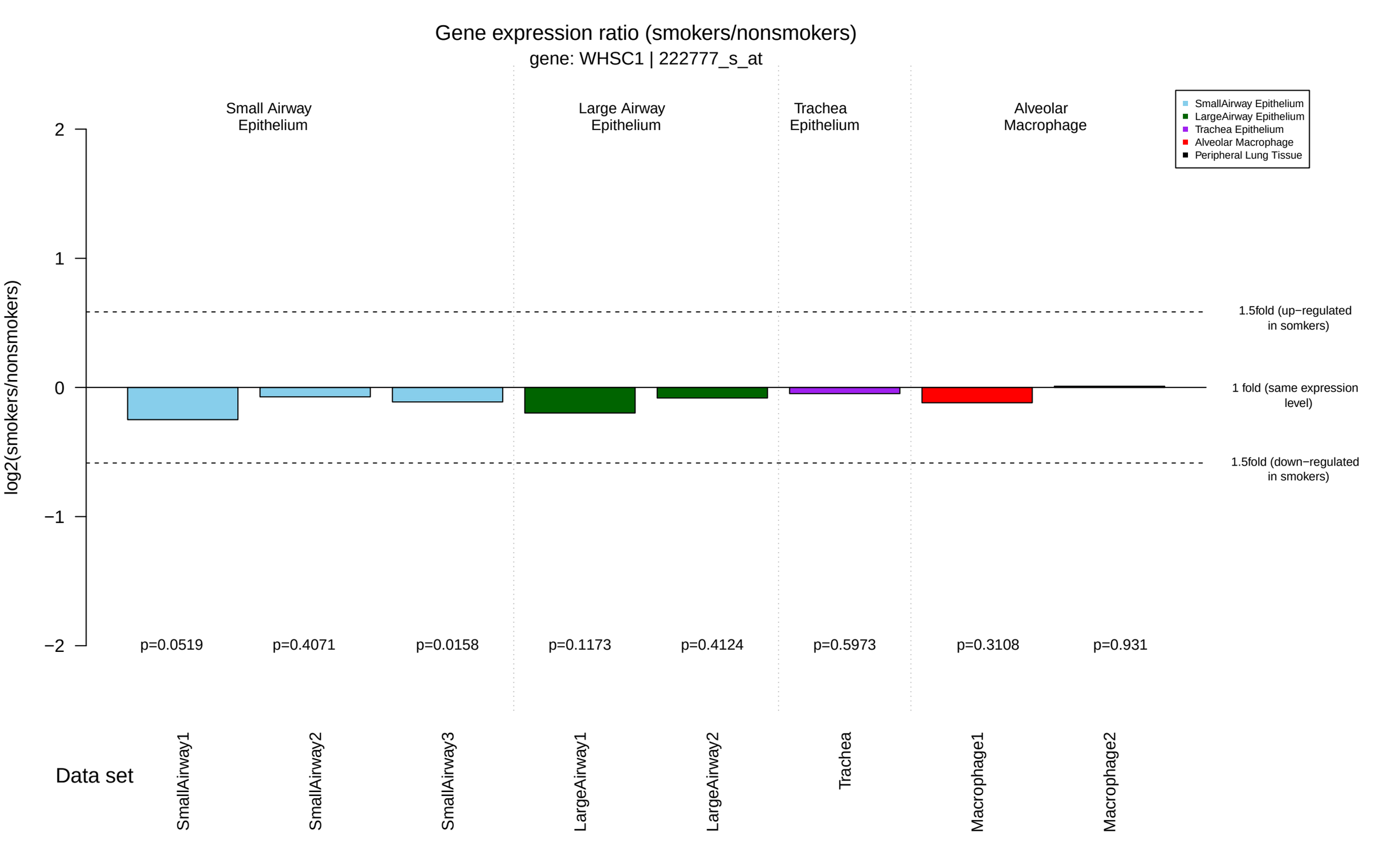
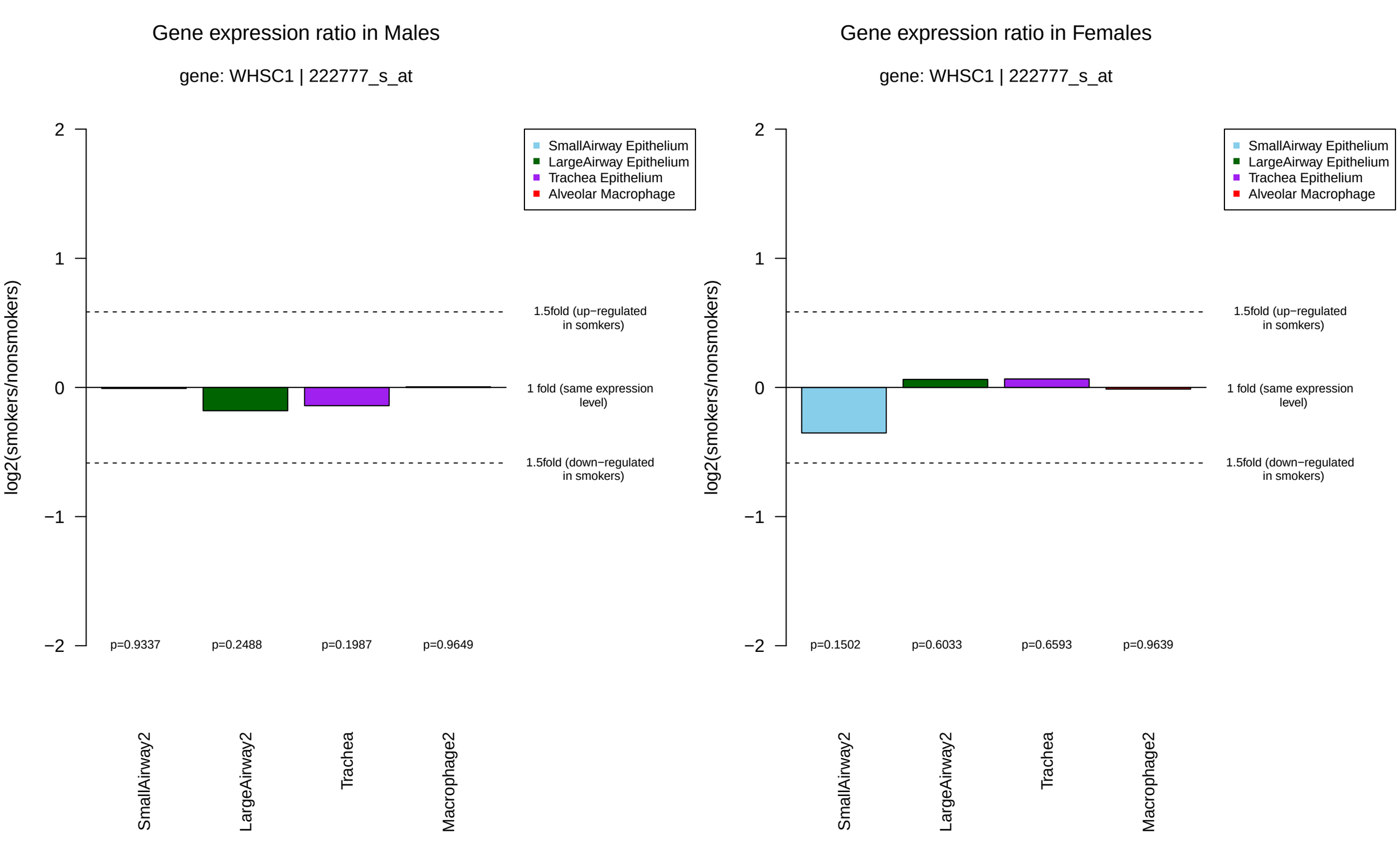
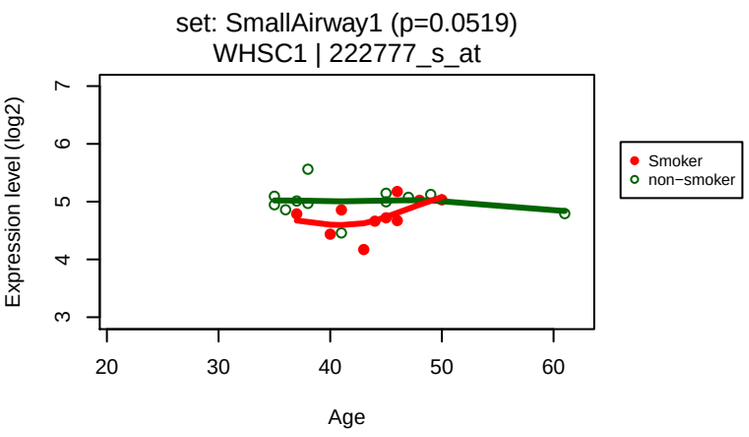
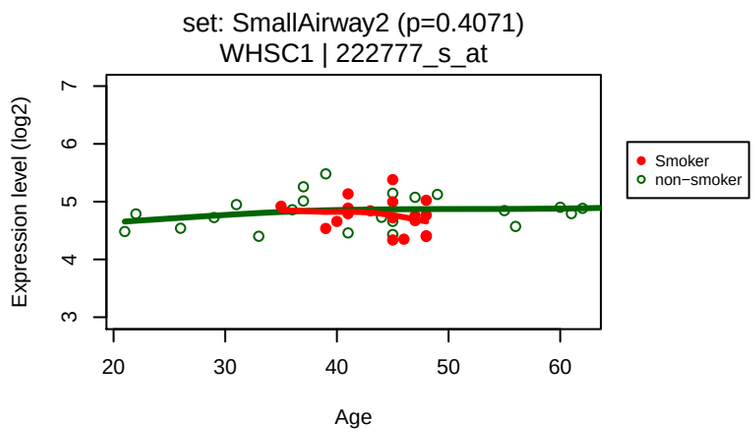
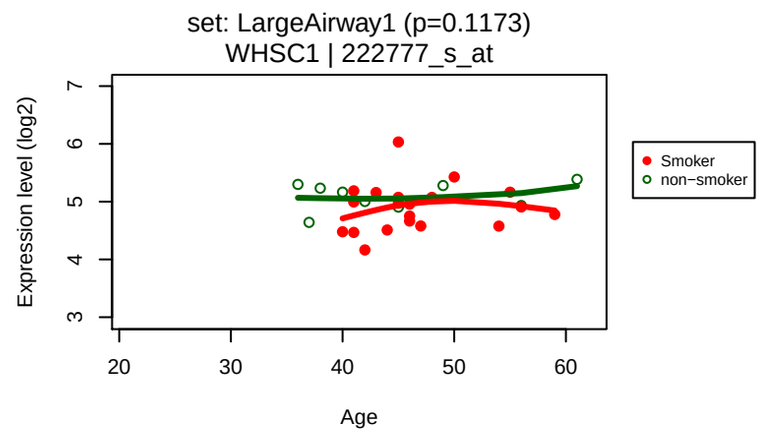

Gene expression of smokers and nonsmokers
gene: WHSC1 | 222777_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0519
p=0.4071
p=0.0158
p=0.1173
p=0.4124
p=0.5973
p=0.3108
p=0.931
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

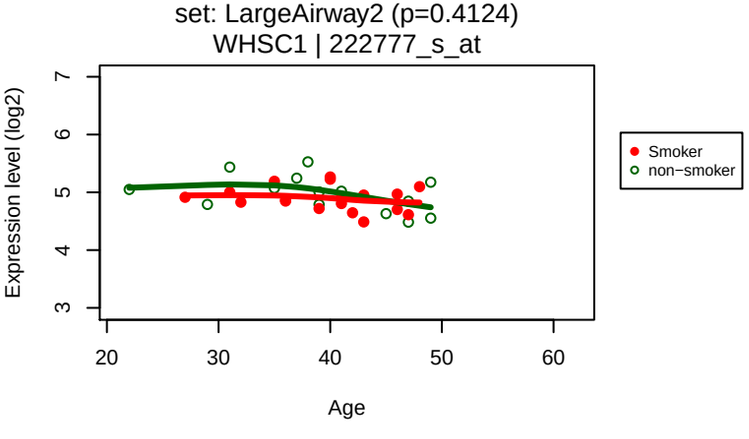

Gene expression ratio (smokers/nonsmokers)
gene: WHSC1 | 222777_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0519
p=0.4071
p=0.0158
p=0.1173
p=0.4124
p=0.5973
p=0.3108
p=0.931
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

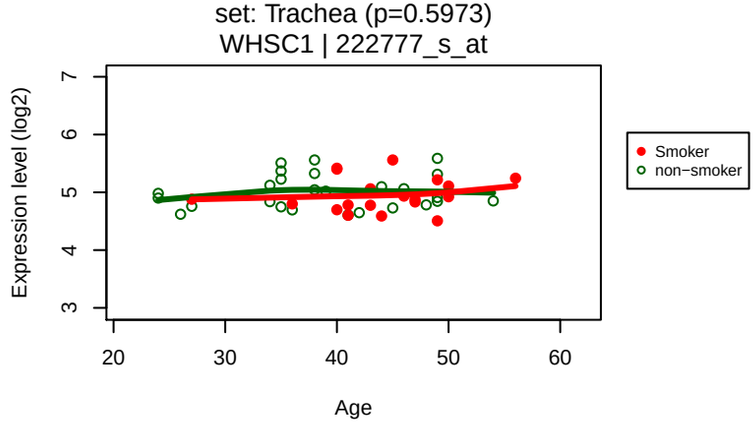

Gene expression ratio in Males
gene: WHSC1 | 222777_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9337
p=0.2488
p=0.1987
p=0.9649
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WHSC1 | 222777_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1502
p=0.6033
p=0.6593
p=0.9639
SmallAirway2
LargeAirway2
Trachea
Macrophage2

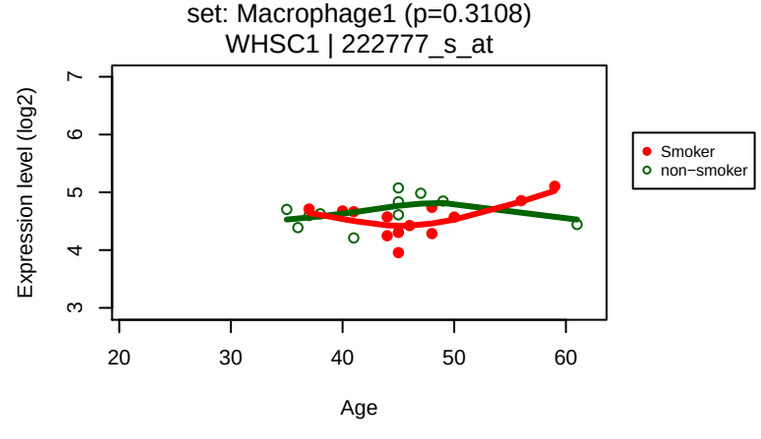

set: SmallAirway1 (p=0.0519)
WHSC1 | 222777_s_at
Expression level (log2)
Age
Smoker
non−smoker

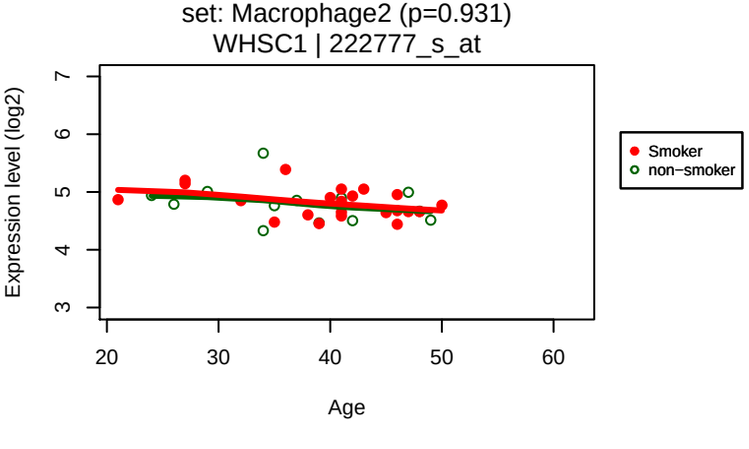

set: SmallAirway2 (p=0.4071)
WHSC1 | 222777_s_at
Expression level (log2)
Age
Smoker
non−smoker

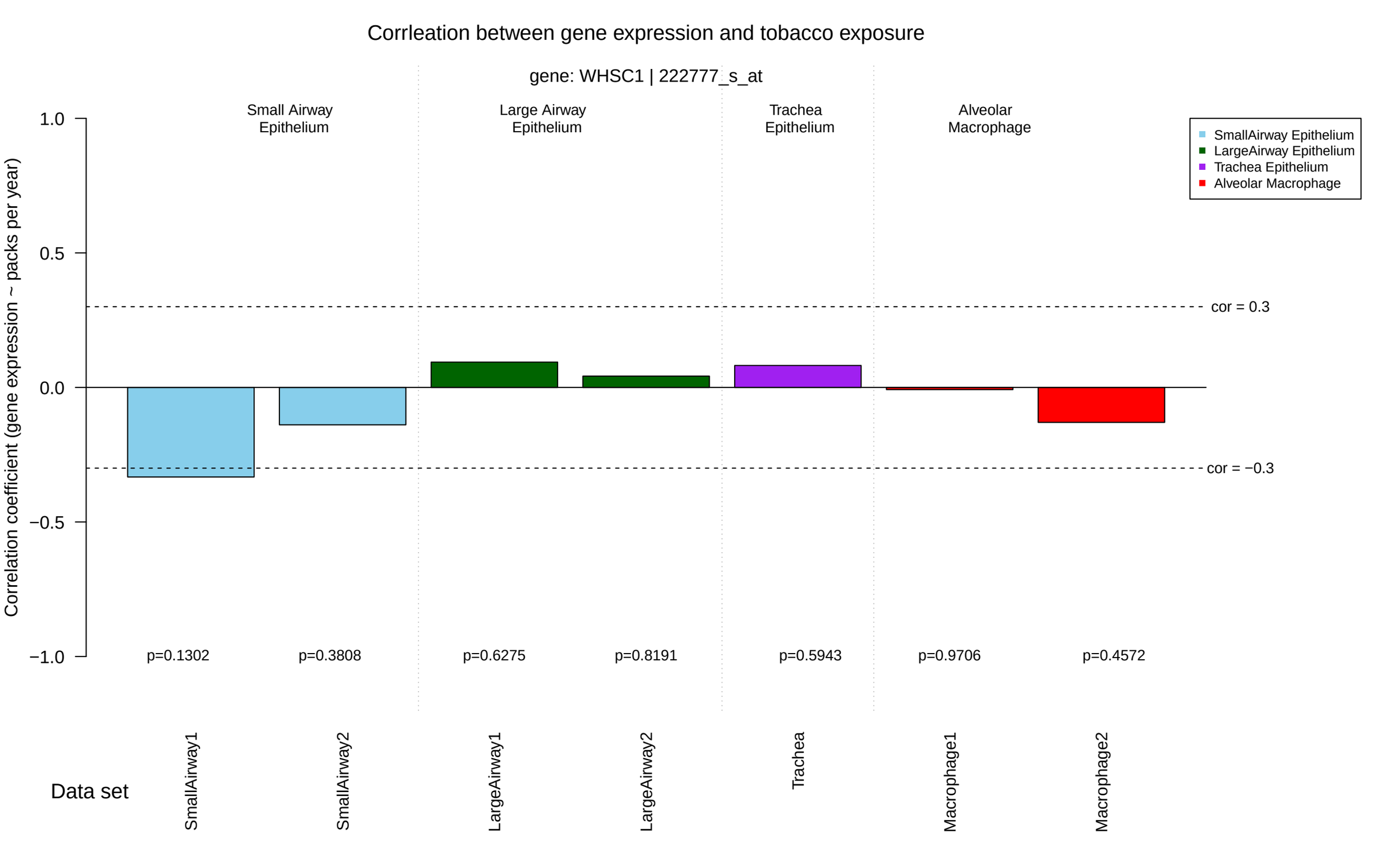

set: LargeAirway1 (p=0.1173)
WHSC1 | 222777_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4124)
WHSC1 | 222777_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5973)
WHSC1 | 222777_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3108)
WHSC1 | 222777_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.931)
WHSC1 | 222777_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WHSC1 | 222777_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1302
p=0.3808
p=0.6275
p=0.8191
p=0.5943
p=0.9706
p=0.4572
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2